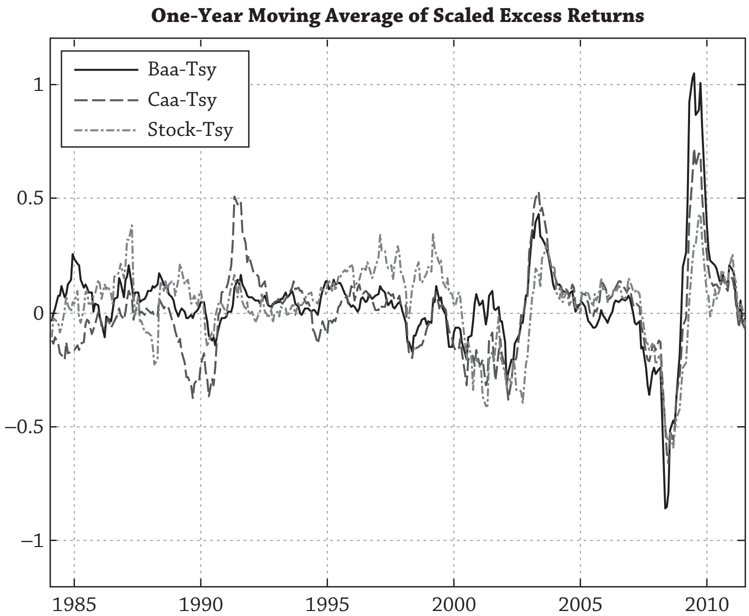

Figure 9.16

and 2001–2001 (dot-com bust). The average credit spread is 1.5%, resulting in a roughly 80 basis point premium for default risk.

The credit spread puzzle is analogous to the equity premium puzzle in chapter 8. New models explain the pronounced difference between credit loss rates and credit spreads by high-and time-varying risk premiums and can jointly match the high equity risk premium and the high credit spread. Corporate bonds and equity must be related as both securities divide up the total value of firms. Corporate bondholders are paid first, while equity holders are residual claimants. Thus, if firm values vary over time, we should expect the changing value of corporate bonds and equity to be linked. Bad times, where firm values shrink enough to jeopardize bondholder claims, result in poor returns for both equity and bonds. (This is the area of $V < 100$ in the example of section 4.2.)

Figure 9.16 plots rolling one-year excess returns over Treasuries for Baa investment grade, Caa high yield, and equities from July 1983 to December 2011. The excess returns have all been scaled so that they have the same volatility as excess equity returns, which is 16.2% in the sample. Figure 9.16 shows a pronounced commonality of corporate bond returns and equity returns; the correlation over the sample between Baa and equities is 48% and the correlation between Caa and equities is 65%. Figure 9.16 shows that after 2005, the co-movement is even higher, particularly during the financial crisis. Both equities and corporate bonds suffered terrific losses during 2008 and 2009 and then bounced back in 2009 and